38.1: Flowers, double fertilization, and fruits are key features of the angiosperm life cycle

The life cycles of all plants are characterized by an alternation of generations, in which multicellular haploid (n) and multicellular diploid (2n) generations alternately produce each other (see Figure 13.6b). The diploid plant, the *sporophyte*, produces haploid spores by meiosis. These spores divide by mitosis, giving rise to multicellular *gametophytes*, the male and female haploid plants that produce gametes (sperm and eggs). Fertilization, the fusion of gametes, results in a diploid zygote, which divides by mitosis and forms a new sporophyte. In angiosperms, the sporophyte is the dominant generation: It is larger, more conspicuous, and longer-lived than the gametophyte. The key traits of the angiosperm life cycle can be remembered as the "three Fs"—*f*lowers, double *f*ertilization, and *f*ruits. We'll begin with flowers.

Flower Structure and Function

In angiosperms, the **flower** is the sporophytic structure specialized for sexual reproduction. A flower is typically composed of four types of floral organs: ****__carpels__, ****__stamens__, ****__petals__, and ****__sepals__ **(Figure 38.2)**. When viewed from above, these organs take the form of concentric whorls. Carpels form the first (innermost) whorl, stamens the second, petals the third, and sepals the fourth (outermost) whorl. All are attached to a part of the stem called the **receptacle**. Flowers are determinate shoots; they cease growing after the flower and fruit are formed.

Figure 38.2

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The structure of an idealized flower.
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An idealized flower has a stamen (anther and filament), single carpel (stigma, style, ovary, and ovule), petals, sepals, and receptacle.

Carpels and stamens are sporophylls—modified leaves specialized for reproduction (see Concept 30.1); sepals and petals are sterile modified leaves. A carpel (megasporophyll) has an **ovary** at its base and a long, slender neck called the **style**. At the top of the style is a sticky structure called the **stigma** that captures pollen. Within the ovary are one or more ****__ovules__, which become seeds if fertilized; the number of ovules depends on the species. The flower in Figure 38.2 has a single carpel, but many species have multiple carpels. In most species, the carpels are fused, resulting in a compound ovary with two or more chambers, each containing one or more ovules. The term **pistil** is sometimes used to refer to a single carpel or two or

more fused carpels **(Figure 38.3)**. A stamen (microsporophyll) consists of a stalk called the filament and a terminal structure called the ****anther**; within the anther are chambers called microsporangia (pollen sacs) that produce pollen. Petals are typically more brightly colored than sepals and advertise the flower to insects and other animal pollinators. Sepals, which enclose and protect unopened floral buds, usually resemble leaves more than the other floral organs do.

Figure 38.3

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The relationship between the terms *carpel* and *pistil*.
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Drawings compare a single unfused carpel (simple pistil), multiple unfused carpels (simple pistils), and two variations of fused carpels (compound pistils).

A simple pistil consists of a single, unfused carpel. A compound pistil consists of two or more fused carpels. Some types of flowers have only a single pistil, while other types have many pistils. In either case, the pistils may be simple or compound.

For both single and multiple unfused carpels, a cross-section of a simple ovary shows a green circle surrounded by a thinner light green coat. At the bottom of the green circle are two ovules. For fused carpels, there are two variations shown: multiple stigmas (one stigma from each carpel) and lobed stigma (one lobe from each carpel). A cross-section of a compound ovary shows one chamber from each carpel, as a green circle divided into thirds like a pie, surrounded by a thin light green coat. Each of the three chambers contains two ovules.

Complete flowers have all four basic floral organs (see Figure 38.2). Some species have **incomplete flowers**, lacking sepals, petals, stamens, or carpels. For example, most grass flowers lack petals. Some incomplete flowers are sterile, lacking functional stamens and carpels; others are *unisexual* (sometimes called *imperfect*), lacking either stamens or carpels. Flowers also vary in size, shape, color, odor, organ arrangement, and time of opening. Some are borne singly, while others are arranged in showy clusters called **inflorescences**. For example, a sunflower consists of a central disk composed of hundreds of tiny incomplete flowers, surrounded by sterile, incomplete flowers that look like yellow petals (see Figure 40.23). Much of floral diversity represents adaptation to specific pollinators.

Methods of Pollination

Pollination is the transfer of pollen to the ovule-bearing structure of a seed plant. In angiosperms, this transfer is from an anther to a stigma. Pollination can occur by wind, water, or animals **(Figure 38.4)**. In wind-pollinated species, including grasses and many trees, the release of clouds of typically small-sized

pollen compensates for the randomness of dispersal by the wind. At certain times of the year, the air is loaded with pollen grains, as anyone plagued with pollen allergies can attest. Some species of aquatic plants rely on water to disperse pollen. Most angiosperm species, however, depend on insects, birds, or other animal pollinators to transfer pollen directly from one flower to another.

Figure 38.4

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Exploring Flower Pollination
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Figure describes different ways flowers can be pollinated: by wind, bees, and moths and butterflies, flies, bats, and birds. Figure describes different ways flowers can be pollinated: by flies, bats, and birds.

Most angiosperm species rely on a living (biotic) or nonliving (abiotic) pollinating agent that can move pollen from the anther of a flower on one plant to the stigma of a flower on another plant. Approximately 80% of all angiosperm pollination is biotic, employing animal go-betweens. Among abiotically pollinated species, 98% rely on wind and 2% on water. (Some angiosperm species can self-pollinate, but such species are limited to inbreeding in nature.)

Abiotic pollination by wind: About 20% of all angiosperm species are wind-pollinated. Since their reproductive success does not depend on attracting pollinators, there has been no selective pressure favoring colorful or scented flowers. Accordingly, the flowers of wind-pollinated species are often small, green, and inconspicuous, and they produce neither scent nor the sugary solution called nectar. Most temperate trees and grasses are wind pollinated. The flowers of hazel (Corylus avellana) and many other temperate, wind-pollinated trees appear in the early spring, when there are no leaves to interfere with pollen movement. The relative inefficiency of wind pollination is compensated for by production of copious amounts of pollen grains. Wind tunnel studies reveal that wind pollination is often more efficient than it appears because floral structures can create eddy currents that aid in pollen capture. Two photos show the hazel carpellate flower (carpels only) and the hazel staminate flowers (stamens only), releasing clouds of pollen.

Pollination by bees: About 65% of all flowering plants require insects for pollination; the percentage is even greater for major crops. Bees are the most important insect pollinators, and there is great concern in Europe and North America that honeybee populations have shrunk. Pollinating bees depend on nectar and pollen for food. Typically, bee-pollinated flowers have a delicate, sweet fragrance. Bees are attracted to bright colors, primarily yellow and blue. Red appears dull to them, but they can see ultraviolet radiation. Many species of bee-pollinated flowers, such as the common dandelion (Taraxacum vulgare), have ultraviolet markings called nectar guides that help insects locate the nectaries (nectar producing glands) but are only visible to human eyes under ultraviolet light. Two photos show that the common dandelion flower under

normal light is yellow. Common dandelion flower under ultraviolet light is red in the center and pale blue on the outside.

Pollination by moths and butterflies: Moths and butterflies detect odors, and the flowers they pollinate are often sweetly fragrant. Butterflies perceive many bright colors, but moth-pollinated flowers are usually white or yellow, which stand out at night when moths are active. A yucca plant (shown here) is typically pollinated by a single species of moth with appendages that pack pollen onto the stigma. The moth then deposits eggs directly into the ovary. The larvae eat some developing seeds, but this cost is outweighed by the benefit of an efficient and reliable pollinator. If a moth deposits too many eggs, the flower aborts and drops off, selecting against individuals that overexploit the plant. The photo shows a moth on a yucca flower.

Pollination by flies: Many fly-pollinated flowers are reddish and fleshy, with an odor like rotten meat. Blowflies visiting carrion flowers (Stapelia species) mistake the flower for a rotting corpse and lay their eggs on it. In the process, the blowflies become dusted with pollen that they carry to other flowers. When the eggs hatch, the larvae find no carrion to eat and die. The photo shows a blowfly sitting on a carrion flower.

Pollination by bats: Bat-pollinated flowers, like moth-pollinated flowers, are light colored and aromatic, attracting their nocturnal pollinators. The lesser long-nosed bat (Leptonycteris curasoae yerbabuenae) feeds on the nectar and pollen of agave and cactus flowers in the southwestern United States and Mexico. In feeding, the bats transfer pollen from plant to plant. Long-nosed bats are an endangered species. Photo shows a long-nosed bat feeding on agave flowers at night.

Pollination by birds: Bird-pollinated flowers, such as columbine flowers, are usually large and bright red or yellow, but they have little odor. Since birds often do not have a well-developed sense of smell, there has been no selective pressure favoring scent production. However, the flowers produce the sugary nectar that helps meet the high energy demands of the pollinating birds. The primary function of nectar, which is produced by nectaries at the base of many flowers, is to “reward” the pollinator. The petals of such flowers are often fused, forming a bent floral tube that fits the curved beak of the bird. The photo shows a humming bird drinking nectar of columbine flower.

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What are the benefits and dangers to a plant of having a highly specific animal pollinator?

*For suggested answer, see*Appendix A*.*

Video

Bee Colony Decline

image

image

EVOLUTION Animal pollinators are drawn to flowers for the food they provide in the form of pollen and nectar. Attracting pollinators that are loyal to a given plant species is an efficient way to ensure that pollen is transferred to another flower of the same species. Natural selection, therefore, favors deviations in floral structure or physiology that make it more likely for a flower to be pollinated regularly by an effective animal species. If a plant species develops traits that make its flowers more prized by pollinators, there is a selective pressure for pollinators to become adept at harvesting food from these flowers. The joint evolution of two interacting species, each in response to selection imposed by the other, is called **coevolution**. For example, some species have fused flower petals that form long, tubelike structures bearing nectaries tucked deep inside. Charles Darwin suggested that coadaptations between flower and insect might lead to correspondences between the length of a floral tube and the length of an insect’s proboscis, a straw-like mouthpart. Based on the length of a long, tubular flower endemic to Madagascar, Darwin predicted the existence of a pollinating moth with a 28-cm-long proboscis. Such a moth was discovered two decades after Darwin’s death **(Figure 38.5)**.

Figure 38.5

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Coevolution of a flower and an insect pollinator.
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A moth’s long proboscis fits into the long floral tube of an orchid.

The long floral tube of the Madagascar orchid *Angraecum sesquipedale* has coevolved with the 28-cm-long proboscis of its pollinator, the hawk moth *Xanthopan morganii praedicta*. The moth is named in honor of Darwin's prediction of its existence.

Climate change may be affecting long-standing relationships between plants and animal pollinators. For example, flowers that require long-tongued pollinators have declined under the warmer conditions in the Rocky Mountains. As a result, there has been selective pressure favoring bumblebees with shorter tongues. Two species of Rocky Mountain bumblebees now have tongues that are about one-quarter shorter than those of bees of the same species 40 years ago.

The Angiosperm Life Cycle: An Overview

Pollination is one step in the angiosperm life cycle. **Figure 38.6** provides a complete overview of the life cycle, focusing on gametophyte development, sperm delivery by pollen tubes, double fertilization, and seed development.

Figure 38.6

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The life cycle of an angiosperm.
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Life cycle of angiosperms including gametophyte development, pollination, double fertilization, and seed development.

For simplicity, a flower with a single carpel (simple pistil) is shown. Many species have multiple carpels, either separate or fused.

Mature flower on sporophyte plant (2 n). Ovary at the base of carpel form ovules with Megasporangium (2 n).

1.In the Megasporangium of each ovule, the megasporocyte divides by meiosis, producing four megaspores. One survives and gives rise to a female gametophyte. Two integuments surround the Megasporangium except at a gap called the micropyle. Inside the female gametophyte (embryo sac) are the antipodal cells, polar nuclei in the central cell, synergids, and egg (n).

2.2. On the anther, each microsporangium (pollen sac) contains microsporocytes (2 n) that divide by meiosis, producing microspores (n).

3.3. A microspore (n) develops into a pollen grain. The generative cell of the gametophyte will divide, forming two sperm. A pollen grain is transferred to a receptive surface of a stigma – the act of pollination.

The tube cell produces the pollen tube (tube nucleus at the tip) and delivers two sperm to the female gametophyte.

4.4. After pollination, eventually two sperm cells (n) are discharged in each ovule.

5.Double fertilization occurs. One sperm (n) fertilizes the egg (n), forming a zygote (2 n). The other sperm fertilizes the central cell, forming the endosperm (a food supply, 3n in this example).

6.The zygote develops into an embryo (2 n) that is packaged along with food into a seed. Fruit tissues would surround the seed but are not described.

7.When a seed germinates, the embryo develops into a mature sporophyte (2 n).

VISUAL SKILLS *Mitosis continues throughout the life cycle. Where in the cycle would the most mitotic divisions occur?*

*For suggested answer, see*Appendix A*.*

Figure 38.6 Walkthrough

The life cycle of an angiosperm

Watch an explanation of this figure in the Figure 38.6 Walkthrough: The life cycle of an angiosperm.

image

image

Animation

Angiosperm Life Cycle

image

image

Animation

Sexual Reproduction in Angiosperms

image

image

Over the course of seed plant evolution, gametophytes became reduced in size and wholly dependent on the sporophyte for nutrients (see Figure 30.2). The gametophytes of angiosperms are the most reduced of all

plants, consisting of only a few cells: They are microscopic, and their development is obscured by protective tissues.

Development of Female Gametophytes (Embryo Sacs)

As a carpel develops, one or more ovules form deep within its ovary, its swollen base. A female gametophyte, also known as an ****__embryo sac__, develops inside each ovule. The process of embryo sac formation occurs in a tissue called the megasporangium **1**within each ovule. Two *integuments* (layers of protective sporophytic tissue that will develop into the seed coat) surround each megasporangium, except at a gap called the *micropyle*. Female gametophyte development begins when one cell in the megasporangium of each ovule, the *megasporocyte,* enlarges and undergoes meiosis, producing four haploid ****__megaspores__. Usually only one megaspore survives; the other three degenerate.

The nucleus of the surviving megaspore divides by mitosis three times without cytokinesis, resulting in one large cell with eight haploid nuclei. The multinucleate mass is then divided by membranes to form the embryo sac. Near the micropyle of the embryo sac, two synergid cells flank the egg and help attract and guide the pollen tube to the embryo sac. At the opposite end of the embryo sac are three antipodal cells of unknown function. The other two nuclei, called polar nuclei, are not partitioned into separate cells but share the cytoplasm of the large central cell of the embryo sac. The mature embryo sac thus consists of eight nuclei contained within seven cells. The ovule, which will become a seed if fertilized, now consists of the embryo sac, enclosed by the megasporangium (which eventually withers) and two surrounding integuments.

Development of Male Gametophytes in Pollen Grains

As the stamens are produced, each anther **2** develops four microsporangia, also called pollen sacs. Within the microsporangia are many diploid cells called *microsporocytes.* Each microsporocyte undergoes meiosis, forming four haploid **microspores**, **3**each of which eventually gives rise to a haploid male gametophyte. Each microspore then undergoes mitosis, producing a haploid male gametophyte consisting of only two cells: the *generative cell* and the *tube cell*. Together, these two cells and the spore wall constitute a ****__pollen grain__. The spore wall, which consists of material produced by both the microspore and the anther, usually exhibits an elaborate pattern unique to the species. During maturation of the male gametophyte, the generative cell passes into the tube cell: The tube cell now has a completely free-standing cell inside it.

Sperm Delivery by Pollen Tubes

After the microsporangium breaks open and releases pollen, a pollen grain may be transferred to a receptive surface of a stigma—the act of pollination. There, it absorbs water and germinates by producing a **pollen tube**, a long cellular protuberance that delivers sperm to the female gametophyte.

A pollen grain typically consists of the spore wall and two cells: a tube cell and a second cell, the generative cell, contained within the tube cell. As the pollen tube elongates through the style, the nucleus of the generative cell divides by mitosis and produces two sperm, which remain inside the tube cell. The tube nucleus leads ahead of the two sperm as the pollen tube grows toward the micropyle in response to chemical attractants produced by the synergid cells. The arrival of the pollen tube initiates the death of one of the two synergids, thereby providing a passageway into the embryo sac. The two sperm are then released from the pollen tube **4** in the vicinity of the female gametophyte.

Double Fertilization

****Fertilization**, the fusion of gametes, occurs after the two sperm reach the female gametophyte. One sperm fertilizes the egg, forming the zygote. The other sperm combines with the two polar nuclei, forming a triploid 3n nucleus in the center of the large central cell of the female gametophyte. This cell will give rise to the **endosperm**, a multicellular, food-storing tissue of the seed. **5** The union of the two sperm cells with different nuclei of the female gametophyte is called ****double fertilization**. Double fertilization ensures that endosperm develops only in ovules where the egg has been fertilized, thereby preventing angiosperms from squandering nutrients on infertile ovules. Near the time of double fertilization, the tube nucleus, the other synergid cell, and the antipodal cells degenerate.

Seed Development

6 After double fertilization, each ovule develops into a seed. Meanwhile, the ovary develops into a fruit, which encloses the seeds and aids in their dispersal by wind or animals. As the sporophyte embryo develops from the zygote, the seed stockpiles proteins, oils, and starch to varying degrees, depending on the species. This is why seeds are such a major nutrient drain. Initially, carbohydrates and other nutrients are stored in the seed's endosperm, but later, depending on the species, the swelling cotyledons (seed leaves) of the embryo may take over this function. When a seed germinates, **7** the embryo develops into a new sporophyte. The mature sporophyte produces its own flowers and fruits: The life cycle is now complete, but it is necessary to examine more closely how an ovule develops into a mature seed.

Seed Development and Structure

After successful pollination and double fertilization, a seed begins to form. During this process, both the endosperm and the embryo develop. When mature, a **seed** consists of a dormant embryo surrounded by

stored food and protective layers.

Endosperm Development

Endosperm usually begins development before the embryo does. After double fertilization, the triploid nucleus of the ovule's central cell divides mitotically, forming a multinucleate "supercell" that has a milky consistency. This liquid mass, the endosperm, becomes multicellular when cytokinesis partitions the cytoplasm by forming membranes between the nuclei. Eventually, these "naked" cells produce cell walls, and the endosperm becomes solid. Coconut "milk" and "meat" are examples of liquid and solid endosperm. The white fluffy part of popcorn is another example of endosperm. The endosperms of just three grains—wheat, maize, and rice—provide much of the food energy for human sustenance.

In grains and most other species of monocots, as well as many eudicots, the endosperm stores nutrients that can be used by the seedling after germination. In other eudicot seeds, the food reserves of the endosperm are completely exported to the cotyledons before the seed completes its development; consequently, the mature seed lacks endosperm.

Embryo Development

The first mitotic division of the zygote is asymmetrical and splits the fertilized egg into a basal cell and a terminal cell **(Figure 38.7)**. The terminal cell eventually gives rise to most of the embryo. The basal cell continues to divide, producing a thread of cells called the suspensor, which anchors the embryo to the parent plant. The suspensor helps in transferring nutrients to the embryo from the parent plant and, in some species, from the endosperm. As the suspensor elongates, it pushes the embryo deeper into the nutritive and protective tissues. Meanwhile, the terminal cell divides several times and then forms a spherical proembryo (early embryo) that is attached to the suspensor. The cotyledons begin to form as bumps on the proembryo. A eudicot embryo, with its two cotyledons, is heart-shaped at this stage of development.

Figure 38.7

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The development of a eudicot plant embryo.
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A eudicot plant embryo develops from the zygote. The eventual seed consists of the seed coat surrounding the two cotyledons, the embryonic root and shoot, the suspensor and the endosperm.

By the time the ovule becomes a mature seed and the integuments harden and thicken into the seed coat, the zygote has given rise to an embryonic plant with rudimentary organs.

Cross-section of a flower showing ovule surrounding the endosperm, endosperm nucleus, zygote and integuments. Zygote is split into a basal cell and terminal cell. The basal cell continues to divide producing the suspensor. The terminal cell also divides and forms the proembryo which is attached to the suspensor. Cotyledons begin to form as bumps on the proembryo forming a heart-shaped eudicot. The seed consists of a seed coat surrounding the two cotyledons; between the two cotyledons is the embryonic shoot apex. At the opposite end of the embryo, where the suspensor attaches, is the embryonic root apex.

Animation

Embryo and Endosperm Development

image

Soon after the rudimentary cotyledons appear, the embryo elongates. Cradled between the two cotyledons is the embryonic shoot apex. At the opposite end of the embryo's axis, where the suspensor attaches, an embryonic root apex forms. After the seed germinates—indeed, for the rest of the plant's life—the apical meristems at the apices of shoots and roots sustain primary growth (see Figure 35.11).

Structure of the Mature Seed

During the last stages of its maturation, the seed dehydrates until its water content is only about 5–15% of its weight. The embryo, which is surrounded by a food supply (cotyledons, endosperm, or both), enters **dormancy**; that is, it stops growing and its metabolism nearly ceases. The embryo and its food supply are enclosed by a hard, protective **seed coat** formed from the integuments of the ovule. In some species, dormancy is imposed by the presence of an intact seed coat rather than by the embryo itself.

If you split apart a seed of the garden bean, a type of eudicot, you can see that the embryo consists of an elongate structure, the embryonic axis, attached to two thick, fleshy cotyledons **(Figure 38.8a)**. Below where the cotyledons are attached, the embryonic axis is called the **hypocotyl** (from the Greek *hypo*, under). The hypocotyl terminates in the **radicle**, or embryonic root. The portion of the embryonic axis above where the cotyledons are attached and below the first pair of miniature leaves is the **epicotyl** (from the Greek *epi*, on, over). The epicotyl, young leaves, and shoot apical meristem are collectively called the *plumule*.

Figure 38.8

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Seed structure.
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Seed structures are compared.

Part a. Common garden bean, a eudicot with thick cotyledons. The fleshy cotyledons store food absorbed from the endosperm before the seed germinates.

Part b. Maize, a monocot. Like all monocots, maize has only one cotyledon. Maize and other grasses have a large cotyledon called a scutellum. The rudimentary shoot is sheathed in a structure called the coleoptile, and the coleorhiza covers the young root.

MAKE CONNECTIONS *In addition to cotyledon number, how do the structures of monocots and eudicots differ? (See*Figure 30.16*.)**

VISUAL SKILLS *Which mature seed lacks an endosperm? What happened to it?*

The cotyledons of the common garden bean are packed with starch before the seed germinates because they absorbed carbohydrates from the endosperm when the seed was developing. However, the seeds of some eudicot species, such as castor beans (*Ricinus communis*), retain their food supply in the endosperm and have very thin cotyledons. The cotyledons absorb nutrients from the endosperm and transfer them to the rest of the embryo when the seed germinates.

The embryos of monocots possess only a single cotyledon **(Figure 38.8b)**. Grasses, including maize and wheat, have a specialized cotyledon called a *scutellum* (from the Latin *scutella*, small shield, a reference to its shape). The scutellum, which has a large surface area, is pressed against the endosperm, from which it absorbs nutrients during germination. The embryo of a grass seed is enclosed within two protective sheathes: a **coleoptile**, which covers the young shoot, and a **coleorhiza**, which covers the young root. Both structures aid in soil penetration during early germination.

Seed Dormancy: An Adaptation for Tough Times

The environmental conditions required to break seed dormancy vary among species. Some seed types germinate as soon as they are in a suitable environment. Others remain dormant, even if sown in a favorable place, until a specific environmental cue causes them to break dormancy.

The requirement for specific cues to break seed dormancy increases the chances that germination will occur at a time and place most advantageous to the seedling. The seeds of many desert plants, for instance, germinate only after a substantial rainfall. If they were to germinate after a mild drizzle, the soil might soon become too dry to support the seedlings. Where natural fires are common, many seeds require intense heat or smoke to break dormancy; seedlings are therefore most abundant after fire has cleared away competing vegetation. Where winters are harsh, seeds may require extended exposure to cold before they germinate; seeds sown during summer or fall will therefore not germinate until the following spring, ensuring a long growth season before the next winter. Many small seeds require light for germination, an adaptation that prevents them from germinating when they are buried so deeply in the soil that their energy reserves would

be exhausted before their plumules could reach sunlight. Some seeds have coats that must be weakened by chemical attack as they pass through an animal's digestive tract and thus are usually carried a long distance before germinating from feces.

The length of time a dormant seed remains viable and capable of germinating varies from a few days to decades or even longer, depending on the plant species and environmental conditions. The oldest carbon-14-dated seed that has grown into a viable plant was a 2,000-year-old date palm seed from Israel. Most seeds are durable enough to last a year or two until conditions are favorable for germinating. Thus, the soil has a bank of ungerminated seeds that may have accumulated for several years. This is one reason vegetation reappears so rapidly after an environmental disruption such as fire.

Sporophyte Development from Seed to Mature Plant

When environmental conditions are conducive for growth, seed dormancy is lost and germination proceeds. Germination is followed by growth of stems, leaves, and roots, and eventually by flowering.

Seed Germination

Seed germination is initiated by **imbibition**, the uptake of water due to the low water potential of the dry seed. Imbibition causes the seed to expand and rupture its coat and triggers changes in the embryo that enable it to resume growth. Following hydration, enzymes digest the storage materials of the endosperm or cotyledons, and the nutrients are transferred to the growing regions of the embryo.

The first organ to emerge from the germinating seed is the radicle, the embryonic root. The development of a root system anchors the seedling in the soil and supplies it with water necessary for cell expansion. A ready supply of water is a prerequisite for the next step, the emergence of the shoot tip into the drier conditions encountered above ground. In garden beans, for example, a hook forms in the hypocotyl, and growth pushes the hook above ground **(Figure 38.9a)**. In response to light, the hypocotyl straightens, the cotyledons separate, and the delicate epicotyl, now exposed, spreads its first true leaves (as distinct from the cotyledons, or seed leaves). These leaves expand, become green, and begin making food by photosynthesis. The shriveled, energy-depleted cotyledons that supplied food for the developing embryo are shed.

Figure 38.9

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Two common types of seed germination.
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Two common types of seed germination are given.

Part a. Common garden bean. In common garden beans, straightening of a hook in the hypocotyl pulls the cotyledons from the soil.

Part b. Maize. In maize and other grasses, the shoot grows straight up through the tube of the coleoptile.

VISUAL SKILLS *How do bean and maize seedlings protect their shoot systems as they push through the soil?*

*For suggested answer, see*Appendix A.**

Animation

Seed Germination

Some monocots, such as maize and other grasses, use a different method for breaking ground as they germinate **(Figure 38.9b)**. The coleoptile pushes up through the soil and into the air. The shoot tip grows through the tunnel provided by the coleoptile and breaks through the coleoptile's tip upon emergence.

Growth and Flowering

Once a seed has germinated and started to photosynthesize, most of the plant's resources are devoted to primary and secondary growth of stems, leaves, and roots (also known as *vegetative growth*). This growth arises from the activity of meristematic cells (see Concept 35.2). During this stage, plants usually photosynthesize and grow as much as possible before flowering, the reproductive phase.

The flowers of a given plant species typically appear suddenly and simultaneously at a specific time of year. Such timing promotes outbreeding, the main advantage of sexual reproduction. Flower formation involves a developmental switch in the shoot apical meristem from a vegetative to a reproductive growth mode. This transition into a *floral meristem* is triggered by a combination of environmental cues (such as day length) and internal signals, as you'll learn in Concept 39.3. Once the transition to flowering has begun, the order of each organ's emergence from the floral meristem determines whether it will develop into a sepal, petal, stamen, or carpel (see Figure 35.33).

Fruit Structure and Function

Before a seed can germinate and develop into a mature plant, it must be deposited in suitable soil. Fruits play a key role in this process. A ****fruit** is the mature ovary of a flower. While the seeds are developing from ovules, the flower develops into a fruit **(Figure 38.10)**. The fruit protects the enclosed seeds and, when mature, aids in their dispersal by wind or animals. Fertilization triggers hormonal changes that cause the ovary to begin its transformation into a fruit. If a flower has not been pollinated, fruit typically does not develop, and the flower usually withers and dies.

Figure 38.10

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The flower-to-fruit transition.
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Fruit forms on American pokeweed.

After flowers are fertilized, stamens and petals fall off, stigmas and styles wither, and the ovary walls that house the developing seeds swell to form fruits. Developing seeds and fruits are major sinks for sugars and other carbohydrates. American pokeweed flowers and fruits are shown here.

During fruit development, the ovary wall becomes the *pericarp*, the thickened wall of the fruit. In some fruits, such as soybean pods, the ovary wall dries out completely at maturity, whereas in other fruits, such as grapes, it remains fleshy. In still others, such as peaches, the inner part of the ovary becomes stony (the pit) while the outer parts stay fleshy. As the ovary grows, the other parts of the flower usually wither and are shed.

Fruits are classified into several types, depending on their developmental origin. Most fruits are derived from a single carpel or several fused carpels and are called **simple fruits(Figure 38.11a)**. An **aggregate fruit** results from a single flower that has more than one separate carpel, each forming a small fruit **(Figure 38.11b)**. These “fruitlets” are clustered together on a single receptacle, as in a raspberry. A **multiple fruit** develops from an inflorescence, a group of flowers tightly clustered together. When the walls of the many ovaries start to thicken, they fuse together and become incorporated into one fruit, as in a pineapple **(Figure 38.11c)**.

Figure 38.11

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Developmental origin of different classes of fruits.
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Developmental origins of different classes of fruits are given.

Part a. Simple fruit. A simple fruit develops from a single carpel (or several fused carpels) of one flower (examples: pea, lemon, peanut).

Part b. Aggregate fruit. An aggregate fruit develops from many separate carpels of one flower (examples: raspberry, blackberry, strawberry).

Part c. Multiple fruit. A multiple fruit develops from many carpels of the many flowers that form an inflorescence (examples: pineapple, fig).

Part d. Accessory fruit. An accessory fruit develops largely from tissues other than the ovary. In the apple fruit, the ovary is embedded in a fleshy receptacle.

In some angiosperms, other floral parts contribute to what we commonly call the fruit. Such fruits are called **accessory fruits**. In apple flowers, the ovary is embedded in the receptacle, and the fleshy part of this simple fruit is derived mainly from the enlarged receptacle; only the apple core develops from the ovary **(Figure 38.11d)**. Another example is the strawberry, an aggregate fruit consisting of an enlarged receptacle studded with tiny, partially embedded fruits, each bearing a single seed.

A fruit usually ripens about the same time that its seeds complete their development. Whereas the ripening of a dry fruit, such as a soybean pod, involves the aging and drying out of fruit tissues, the process in a fleshy fruit is more elaborate. Complex interactions of hormones result in an edible fruit that entices animals that disperse the seeds. The fruit's "pulp" becomes softer as enzymes digest components of cell walls. The color usually changes from green to another color, making the fruit more visible among the leaves. The fruit becomes sweeter as organic acids or starch molecules are converted to sugar, which may reach a concentration of 20% in a ripe fruit. **Figure 38.12** examines some mechanisms of seed and fruit dispersal in more detail.

Figure 38.12

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Exploring Fruit and Seed Dispersal
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The figure presents examples of three major types of fruit and seed dispersal: by water, wind, and animals.

A plant's life depends on finding fertile ground. But a seed that falls and sprouts beneath the parent plant will stand little chance of competing successfully for nutrients. To prosper, seeds must be widely dispersed. Plants use biotic dispersal agents as well as abiotic agents such as water and wind.

Dispersal by water: Some buoyant seeds and fruits can survive months or years at sea. In coconut, the seed embryo and fleshy white meat (endosperm) are within a hard layer (endocarp) surrounded by a thick and buoyant fibrous husk.

Dispersal by wind: With a wingspan of 12 cm, the giant seed of the tropical Asian climbing gourd Alsomitra macrocarpa glides through the air of the rain forest in wide circles when released. The winged fruit of a maple spins like a helicopter blade, slowing descent and increasing the chance of being carried farther by horizontal winds. Tumbleweeds break off at the ground and tumble across the terrain, scattering their seeds. Some seeds and fruits are attached to umbrella-like parachutes that are made of intricately branched hairs

and often produced in puffy clusters. These dandelion seeds (actually one-seeded fruits) are carried aloft by the slightest gust of wind.

Dispersal by animals: The sharp, tack-like spines on the fruits of puncture vine (Tribulus terrestris) can pierce bicycle tires and injure animals, including humans. When these painful tacks are removed and discarded, the seeds are dispersed. Some animals, such as squirrels, hoard seeds or fruits in underground caches. If the animal dies or forgets the cache's location, the buried seeds are well positioned to germinate. Seeds in edible fruits are often dispersed in feces, such as the black bear feces shown here. Such dispersal may carry seeds far from the parent plant. Ants are chemically attracted to seeds with food bodies rich in fatty acids, amino acids, and sugars. The ants carry the seed to their underground nest, where the food body (the lighter-colored portion shown here) is removed and fed to larvae. Due to the seed's size, unwieldy shape, or hard coating, the remainder is usually left intact in the nest, where it germinates.

In this section, you have learned about the key features of sexual reproduction in angiosperms—flowers, double fertilization, and fruits. Next, we'll examine asexual reproduction.